

TVs and electronic sign or billboards, as it does not have any fixed size or resolution, hence limiting restrictions.

3. Transparent OLEDs: These types of OLEDs have transparent substrates, cathode and anode. They can achieve up to 85% transparency as compared to their substrate, when they are turned off. A transparent OLED display allows light to pass in both directions and can be used as either active or passive matrix.

This technology can be used for head-mounted wearables. In the earlier years, Samsung and LG have unveiled a couple of television models in its transparent and mirror OLED displays by the name of Window TVs.

4. Bottom or Top-emitting OLEDs: These types of OLEDs have a substrate which is either opaque or reflective from one end and the light emitted is passed on to the further layers from that side. They are best suited for active-matrix designs as they can be easily integrated with a non-transparent backplane.

5. Foldable OLEDs: FOLEDs have substrates made of very flexible foils or thin sheets of plastics, making them flexible, lightweight and durable. Foldable OLED displays saw an early adoption by the textile industry where it was sewn onto fabrics with an IC, a controller or a remote to display patterns onto the T-shirt. A few of them even had clothes made up of OLED display entirely. Major OLED manufacturers are taking this further to smartphones and television displays to make tomorrow's tech rollable and easily portable. In recent times, Foldable OLEDs have now been started using in Curved TVs, smartphones like the LG G Flex and the ReFlex, flaunt a gorgeous flexible OLED display.

6. White OLEDs: White OLEDs emit more than usual white light which is brighter and more uniform than the usual fluorescent lights. White OLEDs are known to have true-colour qualities of incandescent lighting, making them energy efficient and are still struggling to find the best use.

OLED displays can be built and used in various ways, considering the different types it has to offer, along with an array of materials. Large displays differ in the way pixels are built or positioned. In some, the red, green, and blue sub-pixels are arranged side by side whereas, in others, the pixels are